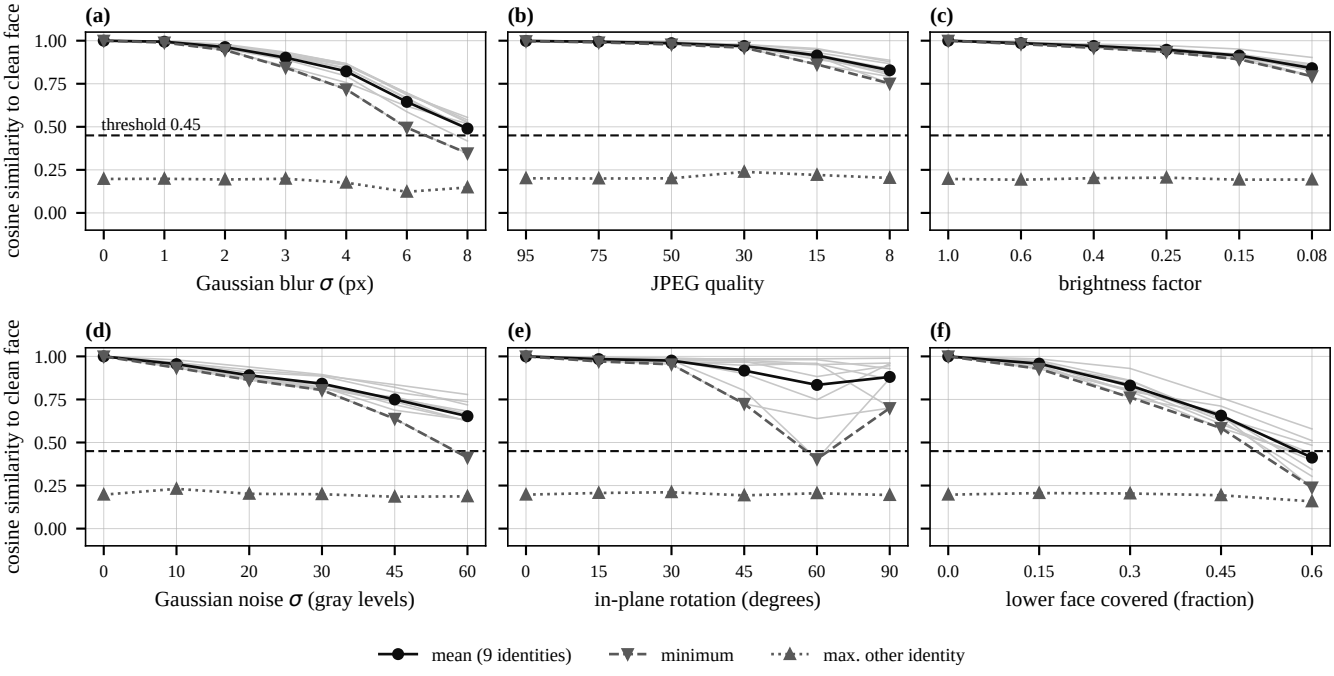
(a)
(b)
(c)
(d)
(e)
(f)
cosine similarity to clean face
threshold 0.45
1.00
0.75
0.50
0.25
0.00
0 1 2 3 4 6 8
Gaussian blur σ (px)
95 75 50 30 15 8
JPEG quality
1.0 0.6 0.4 0.25 0.15 0.08
brightness factor
0 10 20 30 45 60
Gaussian noise σ (gray levels)
0 15 30 45 60 90
in-plane rotation (degrees)
0.0 0.15 0.3 0.45 0.6
lower face covered (fraction)
mean (9 identities)
minimum
max. other identity